(PROJECT PROPSAL)

INTELLIGENT ELECTRIC POLE

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NATTAWAN SAETANG
PIMCHANOK SARARIT

BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY
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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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Abstract

The project proposes the development of a simple electric pole into a smart utility pole that can measure PM2.5 dust or even monitor the weather and humidity in each area and send the results to an app or web so that people wishing know how much pm dust in the area and the weather conditions that day and will be better prepared to go out on a daily basis First of all, our group has been looking for information about smart electric poles for quite some time that in Thailand there are no poles that can measure PM2.5 dust and weather conditions, or if there is, it's not enough and still can't access it. communities or even educational places such as schools or universities. We therefore take the ideas of various research that we have to find information and adapt them to our project to suit the location we want to install our light poles and friends to make people use it more comfortable in life

Keyword: measure dust pm2.5, check the weather, moisture meter

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

As our group studied computer engineering, which learned many areas such as coding and circuitry, our group understood the basic principles of coding and circuitry. Therefore, our group came up with an idea and came up with an idea from the fact that we have seen the light poles of foreign countries and the idea of developing electric poles in Thailand because we are studying at Mae Fah Luang University Chiang Rai province and they have problems with the pm dust value that is quite high, so they think to make a solar light pole or a smart electric pole that can measure the pm dust in each area that the area we live in how much dust is there and how much is it harmful to the body, but electric poles that we can't only measure pm dust but it can also measure the weather in each area as well

1.2 Objective

1.2.1 light up the night

1.2.2 Increase safety in the area where the lamp post is installed.

1.2.3 Use the user to know the weather, dust values in that area where the pole is installed.

1.3 Scope

1.3.1 General user Students and personnel in educational institutions

1) Able to collect weather conditions (temperature, humidity, pm2.5, light intensity)

2) There is an emergency switch to notify security personnel through LINE.

3) show website information

4) Can be a student charging point

1.3.2 Administrator

1) List all admin functions

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.1.1 Find the topic of the project and present it to the advisor and committee.

1.4.2 Literature review and requirement gathering

1.4.2.1 Find related research to compare with the topic the group has chosen and analyze their advantages and disadvantages.

1.4.3 Acquiring system requirements with potential users, i.e., university

1.4.3.1 Survey the university area to collect data for weather analysis and pm dust values in order to install power pole.

1.4.4 System design includes a weather and pm dust senser module

1.4.4.1 Design a system that detects dust, pm and weather conditions

1.4.5 Construction and development of electric poles

1.4.5.1 Follow the steps planned in 1.5.4 test and collect data from power poles making corrections and upgrades more defficient.

1.4.5.2 Repeat the test to make sure the pole can be used.

1.4.6 System evaluation (functionality and user assessment)

1.4.6.1 Finally taken from users to test the functionality and suggestions as well as comments user comments about this project.

1.4.7 System manual

1.4.7.1 Detailed instruction manual for user convenience

1.5 Plan

Table 0.1 Working plan

	Pre-project					Project				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Finding the topic and Proposal preparation										
Research the information										
System analysis										
System design										
Prepare for final presentation										
System development										
Check and solve problem										
Finish software / hardware										
Documentation / Report										
Final presentation										

1.6 Expected Result

1.6.1 Can measure pm2.5

1.6.2 Users can check the dust and weather conditions in the installation area.

1.6.3 The device can work normally.

1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware (Identify also the minimum specification)

- LED sportlight
- Sola cell
- Adapter
- Step up down
- ESP32
- DHT22
- Pms 7003G7
- GPS neo-6m
- UV sensor BH1750

1.7.1.2 Software (List all target software and version)

- DB/firebase
- Webapp/app

1.7.2 Place (Locate the working place such as MFU and/or other places for getting requirements, testing, and installing the system)

1.7.3 Budget
- Sola cell (500THB)
- Adapter (100THB)
- LED sport light (1500THB)
- ESP32 (111THB)
- DHT22 (100THB)
- PMS7003G7 (550THB)
- GPS neo-6m (122THB)
- UV sensor BH1750 (60THB)

= 3100 THB

CHAPTER 2

LITERATURE REVIEW

2.1Related Theory

2.1.1 Entropy theory

- Entropy is used to show the uniformity of many types of energy distribution in space. distribution of pollutants No scholar has used entropy to determine the uniformity of the distribution of pollutant gases or small pollutants. Pollution can be considered as a type of diffuse energy, so this research entropy was used. creatively with the distribution of PM2.5 to study the spread of PM2.5 [1]

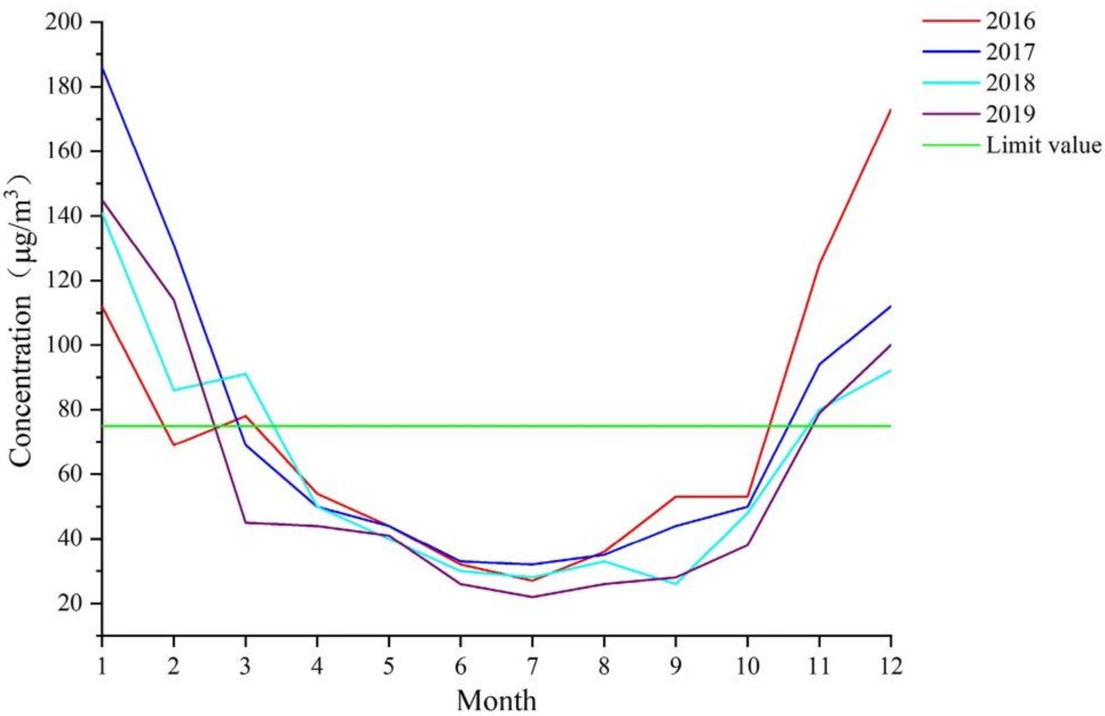

Figure 2.1.1 PM2.5 concentration by month in 2016–2019.

The daily PM2.5 concentration and wind speed at corresponding was plotted in Fig 2.2 The correlation coefficient between them is −0.26757, being weak negative. The PM2.5 concentration is related to the wind speed, but many factors such as vehicle, life and industry would be influential.

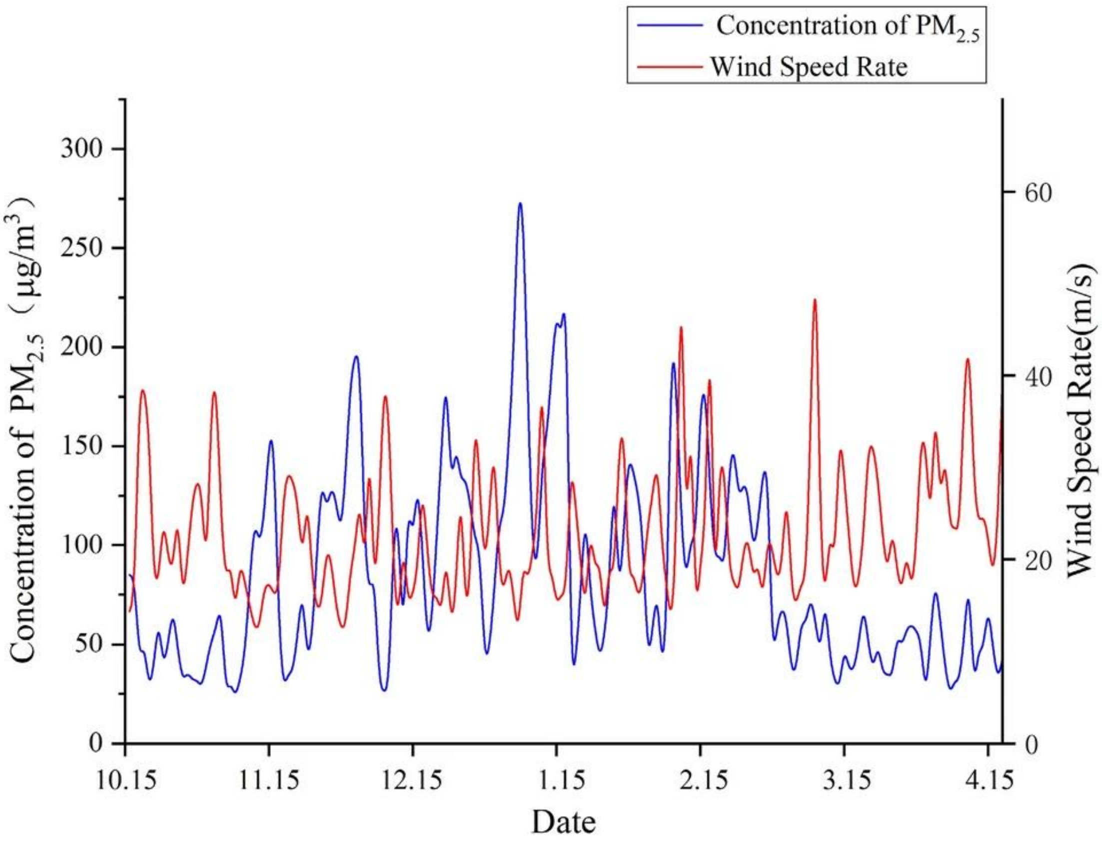

Figure 2.1.2 concentration and wind speed by day the daily PM2.5 concentration and wind speed at corresponding

2.2Related Research

2.2.1 This research is different from the project we are going to do. This article will focus on important infrastructure in modern cities. Pedestrian roads occupy a significant part of the city and absorb abundant solar radiation, so sidewalks have enormous potential for solar energy. and can act as a distributed energy generator in a smart and sustainable city. different from our project Ours will use solar energy from the use of solar panels. as energy storage and used throughout the operation as it is stored [2]

2.2.2This research is different from the project we are going to do. This article will focus on Changing market demand for more commercial use of solar energy Plus, solar energy has the benefit of having unlimited energy. different from our project by Solar energy is the energy that reduces costs and reduces the cost of use, saving the budget for this project, as well as saving energy from electricity caused by global warming. and used throughout the operation as it is stored [3]

2.3 Related Work

2.3.1 A solar fan with a smart electric pole by Saturn Electric Energy Intelligent will have an embedded system This makes it possible to measure weather conditions, p.m2.5 dust values, and UV light intensity measurements. different from a Solar fan because it is an electrical appliance that changes the power source from using electricity to using only solar energy.

2.3.2 How is a solar cell fountain different from our project? The solar cell fountain will have a motor connected to the solar cell and store energy to make electricity to supply the motor by sucking water up into the fountain itself. and our project There will be an embedded system that allows more functions to choose from, such as p.m. 2.5 dust measurement and weather indication.

2.4 Related Technology

2.4.1 Hardware

2.4.1.1 LED sport light

Figure 2.1.3 LED sport light 100W

- Street light LED Street light 100W is a LED street lamp that provides adequate illumination that can be installed in a parking lot. entrance gate and has a long service life Can be installed to various types of street light poles as needed. Can be used with various types of light poles, street light poles as needed

2.4.1.2 Sola cell

Figure 2.1.4 Solar cell 250W

- An electrical device that converts light energy or photons into electrical energy. The resistance, voltage and current values will change when the incident light does not require an external power supply. and when connecting the lamp will cause current to flow through the tube.

2.4.1.3 ESP32

Figure 2.1.5 ESP32 NodeMCU ESP-WROOM-32 Wi-Fi and Bluetooth

- It's the name of a microcontroller IC that supports built-in WiFi and Bluetooth connectivity.

2.4.1.4 DHT22

Figure 2.1.6 DHT22 / AM2302 temperature and humidity sensor

- It is a high-precision humidity and temperature sensor. This sensor measures the relative humidity. A capacitive sensor element is used to measure humidity.

2.4.1.5 Pms 7003G7

Figure 2.1.7 PMS7003 Laser Dust Sensor module measures PM2.5 dust

- It is a dust sensor that can measure PM1.0, PM2.5 and PM10. The sensor measures dust particles as small as 0.3 micrometers. The sensor uses laser light to measure dust in the air, making it more accurate. dust measurement

2.4.1.6 GPS neo-6m

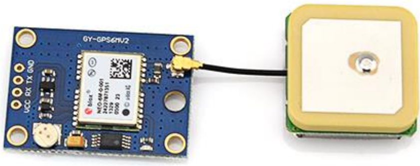

Figure 2.1.8 GY-NEO6MV2 Ublox GPS Module GPS Module

- Ublox NEO-6M GPS Module is a GPS module used to locate various locations in the world. There is a library available.

2.4.1.7 UV sensor BH1750

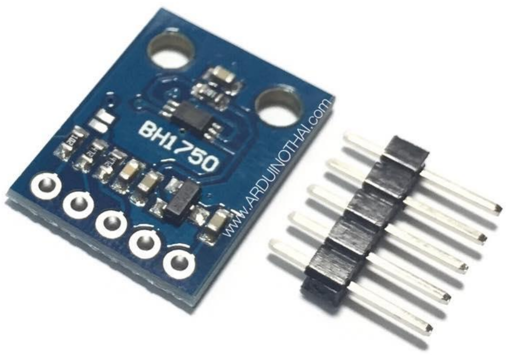

Figure 2.1.9 Light Sensor Module (BH1750)

- The ambient light sensor consists of a photodiode that detects light and converts it into electricity. Metering depends on light intensity. From the block diagram, the PD is a photodiode used to detect light. Its response is approximately the same as the response of the human eye.

2.4.2 Software

2.4.2.1 DB/firebase

Figure 2.2.1 Firebase Realtime Database

- Firebase is a service of Google. It is an online data service in the form of real time database for applications and web applications. A by-product of IoT (Internet of Things) is that Firebase can be used as a medium to connect devices together.

2.4.2.2 Webapp

Figure 2.2.2 Web Application

- Apps that are written to be able to be launched directly in the Web Browser without having to load the full application onto the device, resulting in overall relatively low resource consumption. can be activated quickly And of course, within the Web Application is often optimized to run faster than opening a normal application.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Existing System

The existing could be demonstrated by a use case diagram in **Figure 3.1**

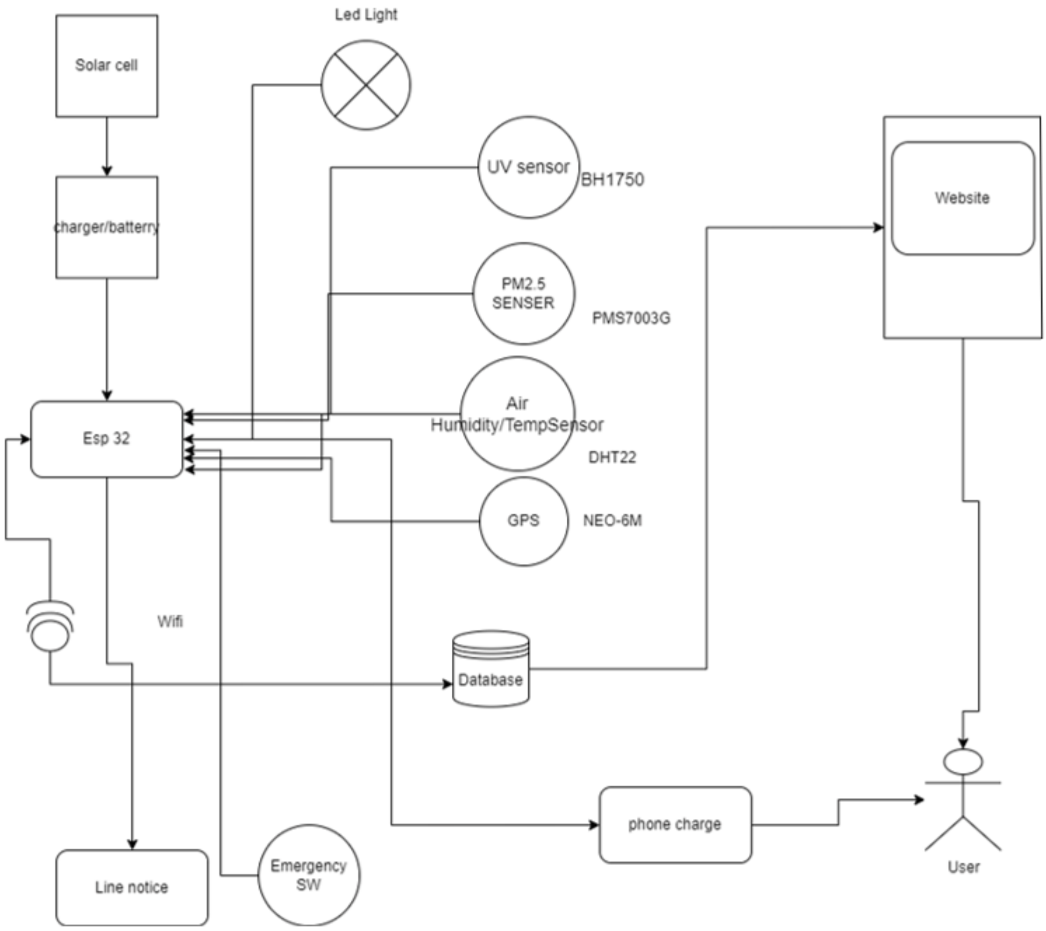

Figure 3.1 Use case diagram of the existing system

3.2 New System

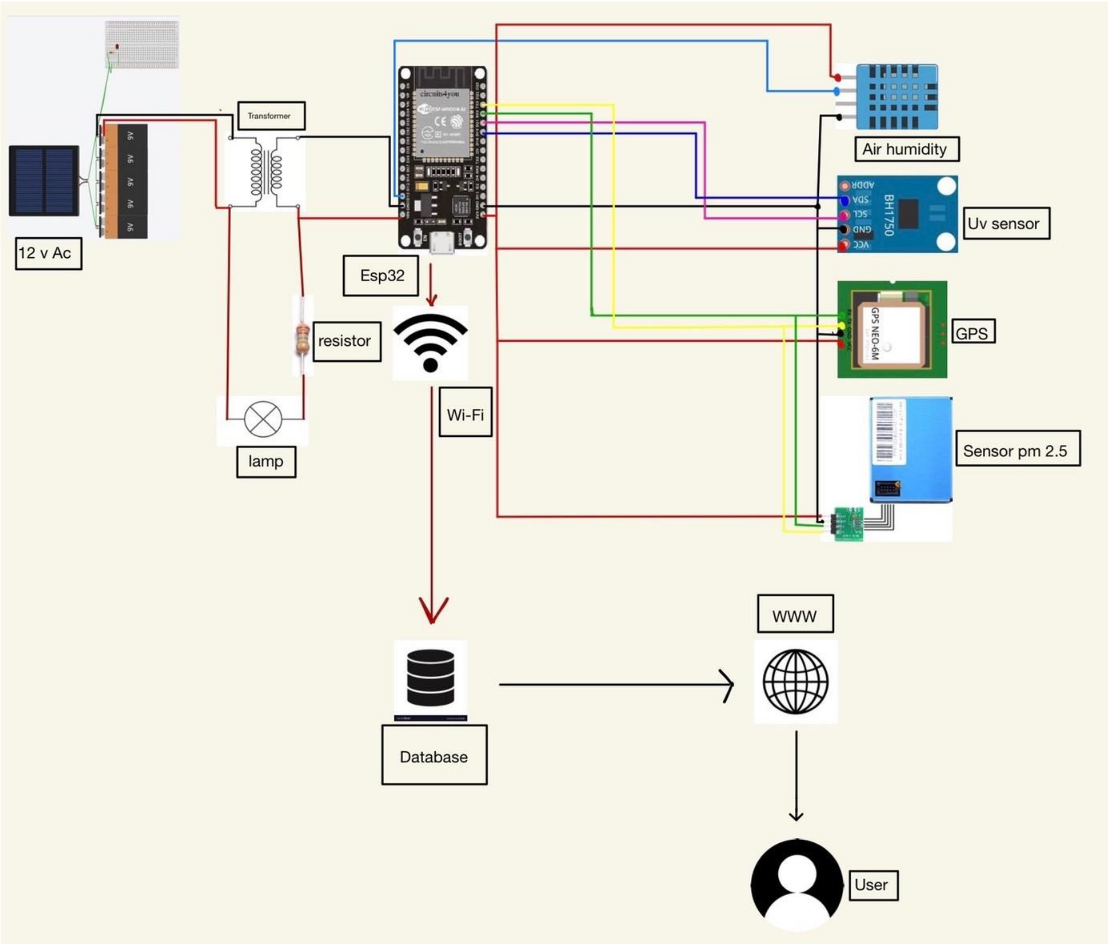

Figure 3.2 Use case diagram of the existing system

3.3 Use case diagram

After gathering user requirements, the new system could be represented by a use case diagram as shown in **Figure 3.2**

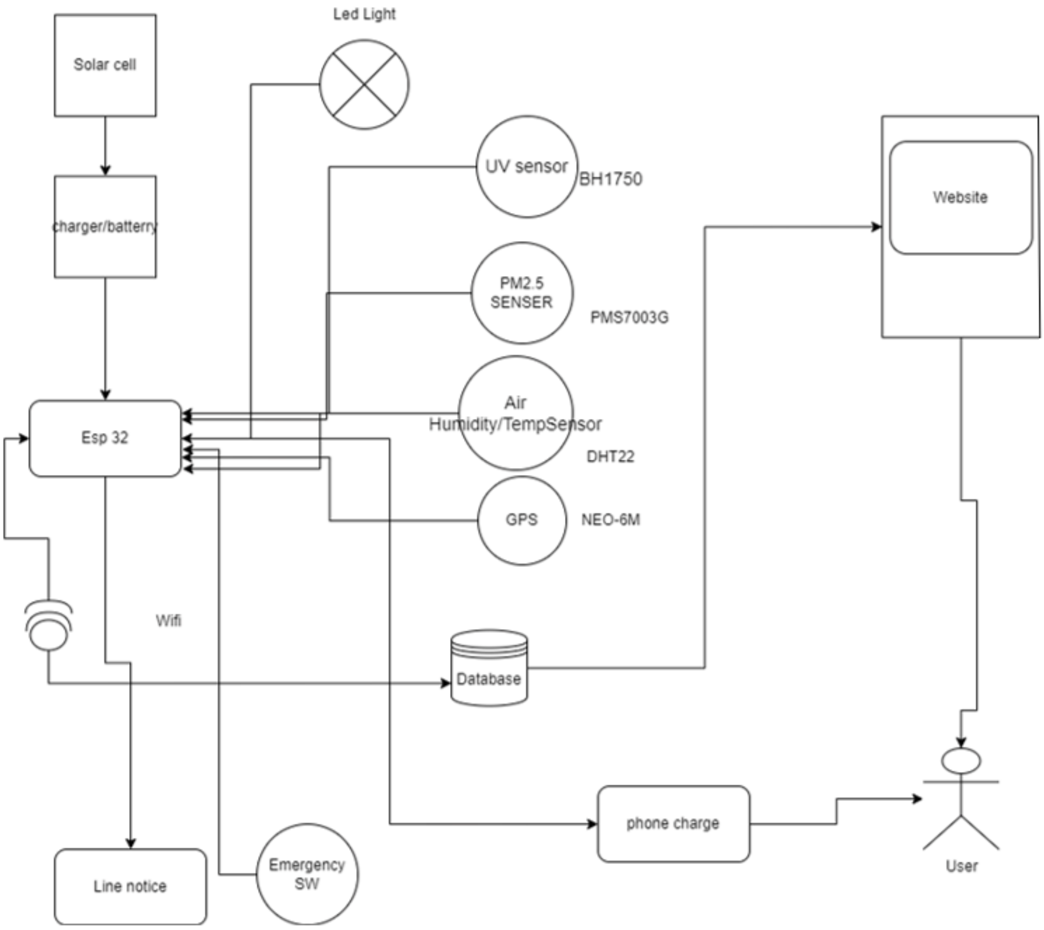

Figure 3.3 Use case diagram of the new system

3.4 Activity diagram

Analysis of a new system derives an activity diagram as shown in **Figure 3.3**

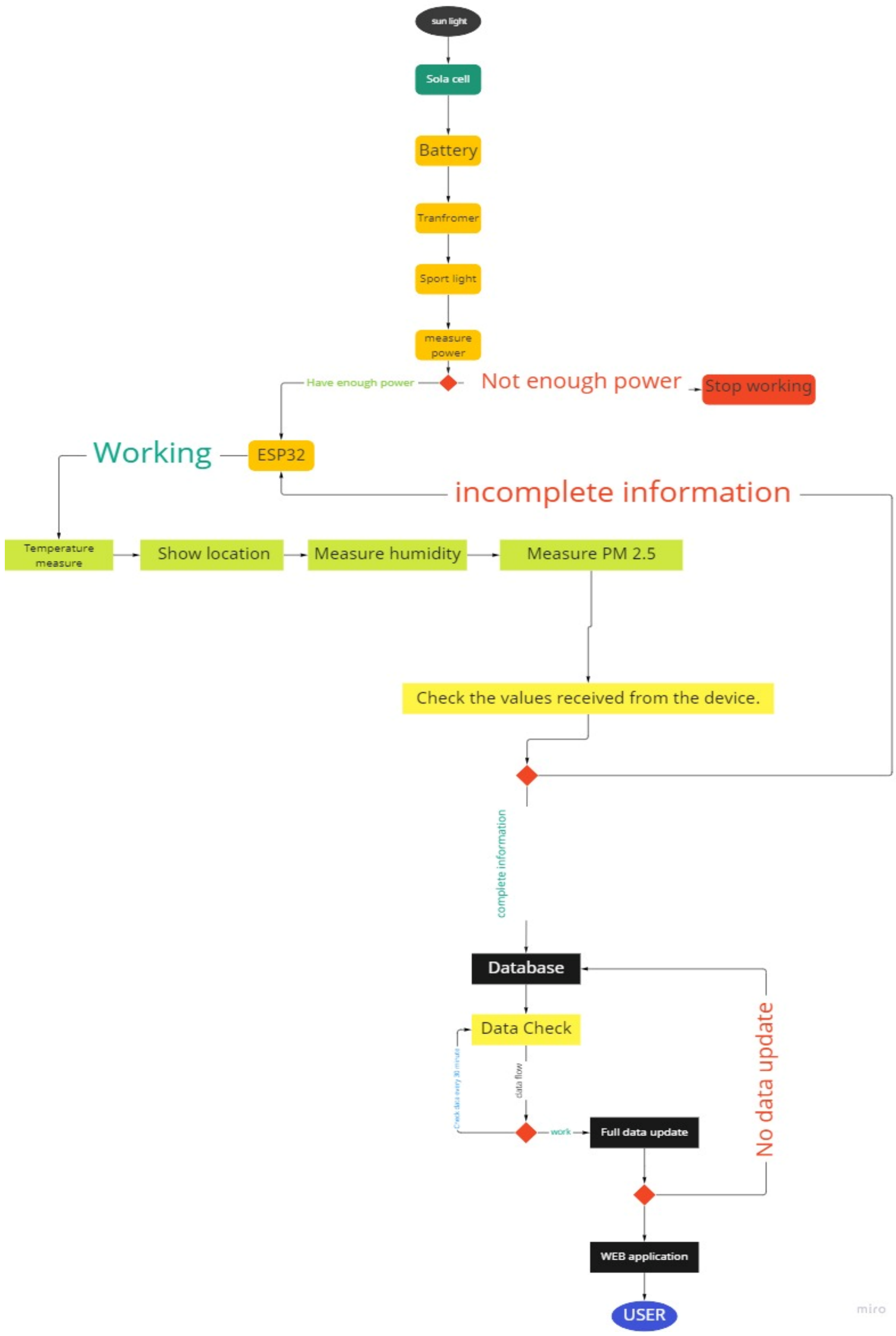

Figure 3.4 Activity diagram of the new system

CHAPTER 4

CONCLUSION

4.1 Conclusion

Summary of making a project for a smart electric pole that can tell PM2.5 dust and weather conditions at different times of the day or even tell the humidity in the air and that we made a project for smart electric poles to bring to our main target groups, schools, or communities. In order to be comfortable and safe to go out to everyday life more and in order to reduce the risk of crime in that place because of the community or educational institutions may not have enough lights in the area, so we want to install our light poles in the locations in our target audience and most importantly, our poles are not just monitoring PM2.5 dust or can only see the weather conditions each day Since our utility poles are powered by solar energy, we do not have to worry about being down or having as many failures as conventional poles.

4.2 Discussion

From the fact that our group has done a project on smart electric poles, the results can be discussed as follows the idea of making a smart electric pole or a power pole that can measure pm2.5 dust and check the weather or humidity every day is what we have learned and have knowledge of coding together. With at present, smart electric poles can be seen in many foreign countries, but in Thailand, they are very few and still do not reach our main target groups, which are educational institutions or communities Therefore, our group has chosen to make this smart electric pole in order to make everyone have a more comfortable life and also for people who are interested in making a smart electric pole our group model has led to improvements and improvements.

This project has received advice and assistance from many teachers in order to benefit those who need it and to make everyone feel more comfortable in life and to be a source of learning for others. For those who are interested in this project or who are thinking of making a smart electric pole as well, they have been able to improve and develop better, which this project has successfully fulfilled the objectives we set.

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